

Fluid Flow Analysis over Aircraft Wing

Low Reynolds Number — 2D Panel Method & XFLR5

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Code: <https://github.com/R-Abinav/airfoil-sim>

Tools: Rust (custom panel method solver), XFLR5 v6.62 (XFoil-based viscous analysis)

1 Introduction

This report analyzes 2D aerodynamic flow over four airfoils at low Reynolds number ($Re = 500,000$): CLARK Y, E387, NACA 2414, and NACA 4412.

Two approaches are used:

1. A custom **vortex panel method** implemented in Rust — inviscid, ideal flow.
2. **XFLR5** using XFOIL's viscous boundary layer model — captures stall and drag.

Airfoil coordinates are sourced from the UIUC Airfoil Database and the simulation sweeps angle of attack α from -5° to 15° .

2 Panel Method — Implementation

2.1 Governing Equation

At low Reynolds number, assuming incompressible, inviscid, irrotational flow, the velocity field satisfies Laplace's equation:

$$\nabla^2 \phi = 0 \tag{1}$$

where ϕ is the velocity potential. Since this is linear, solutions can be superposed.

2.2 Vortex Sheet Discretization

The airfoil surface is divided into N straight panels. Each panel j carries a constant vortex sheet strength γ_j . The freestream is:

$$\vec{V}_\infty = V_\infty (\cos \alpha, \sin \alpha) \tag{2}$$

Panel geometry — for consecutive points (x_j, y_j) and (x_{j+1}, y_{j+1}) :

$$x_{mid} = \frac{x_j + x_{j+1}}{2}, \quad y_{mid} = \frac{y_j + y_{j+1}}{2} \quad (3)$$

$$\Delta s_j = \sqrt{(x_{j+1} - x_j)^2 + (y_{j+1} - y_j)^2} \quad (4)$$

$$\phi_j = \arctan 2(y_{j+1} - y_j, x_{j+1} - x_j) \quad (5)$$

$$\hat{n}_j = (-\sin \phi_j, \cos \phi_j) \quad (6)$$

2.3 Boundary Condition

No flow through the surface — normal velocity component is zero at each control point i :

$$\vec{V} \cdot \hat{n}_i = 0 \quad \forall i \quad (7)$$

2.4 Influence Coefficients

Panel j induces a normal velocity at control point i . Transforming into panel j 's local frame:

$$X = (x_{mid,i} - x_{j,1}) \cos \phi_j + (y_{mid,i} - y_{j,1}) \sin \phi_j \quad (8)$$

$$Y = -(x_{mid,i} - x_{j,1}) \sin \phi_j + (y_{mid,i} - y_{j,1}) \cos \phi_j \quad (9)$$

Distances and angles to panel endpoints:

$$r_1 = \sqrt{X^2 + Y^2}, \quad r_2 = \sqrt{(X - \Delta s_j)^2 + Y^2} \quad (10)$$

$$\beta_1 = \arctan 2(Y, X), \quad \beta_2 = \arctan 2(Y, X - \Delta s_j), \quad \Delta \beta = \beta_2 - \beta_1 \quad (11)$$

Normal and tangential influence coefficients:

$$A_{ij} = \frac{1}{2\pi} \left[\sin(\phi_i - \phi_j) \ln \frac{r_1}{r_2} + \cos(\phi_i - \phi_j) \Delta \beta \right] \quad (12)$$

$$B_{ij} = \frac{1}{2\pi} \left[\cos(\phi_i - \phi_j) \ln \frac{r_1}{r_2} - \sin(\phi_i - \phi_j) \Delta \beta \right] \quad (13)$$

Self-influence: $A_{ii} = \frac{1}{2}$, $B_{ii} = 0$.

2.5 Linear System

Enforcing the boundary condition at all N control points:

$$\mathbf{A}\boldsymbol{\gamma} = \mathbf{b}, \quad b_i = V_\infty \sin(\phi_i - \alpha) \quad (14)$$

2.6 Kutta Condition

Flow must leave the trailing edge smoothly. The last row of $\mathbf{A}$ is replaced by:

$$\gamma_1 + \gamma_N = 0 \quad (15)$$

The system is solved via LU decomposition.

2.7 Output Quantities

Tangential velocity at panel i :

$$V_{t,i} = V_\infty \cos(\phi_i - \alpha) + \sum_{j=1}^N B_{ij} \gamma_j \quad (16)$$

Pressure coefficient:

$$C_{p,i} = 1 - \left(\frac{V_{t,i}}{V_\infty} \right)^2 \quad (17)$$

Lift coefficient (Kutta–Joukowski, chord $c = 1$):

$$C_l = \frac{2}{V_\infty} \sum_{j=1}^N \gamma_j \Delta s_j \quad (18)$$

3 Panel Method Results

3.1 Lift Coefficient vs Angle of Attack

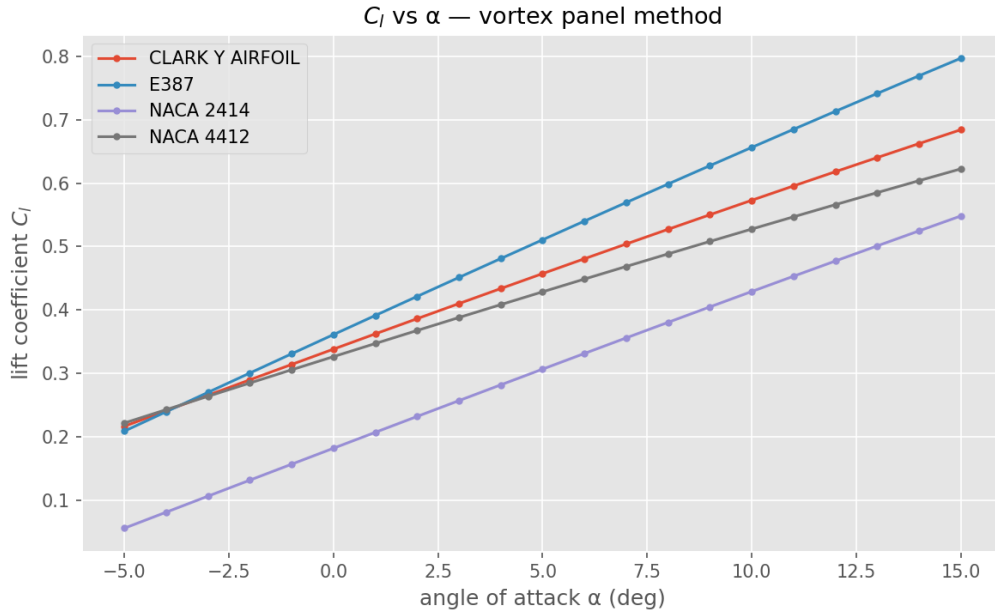


Figure 1: C_l vs α for all four airfoils — vortex panel method. Linear slope ≈ 0.10 – 0.11 /rad consistent with thin airfoil theory (2π).

Airfoil	Format	Panels	C_l at 0°	$dC_l/d\alpha$
CLARK Y	Lednicer	120	0.338	0.024/deg
E387	Selig	60	0.361	0.031/deg
NACA 2414	Selig	60	0.182	0.025/deg
NACA 4412	Selig	34	0.326	0.021/deg

Table 1: Panel method results summary.

3.2 Pressure Distribution

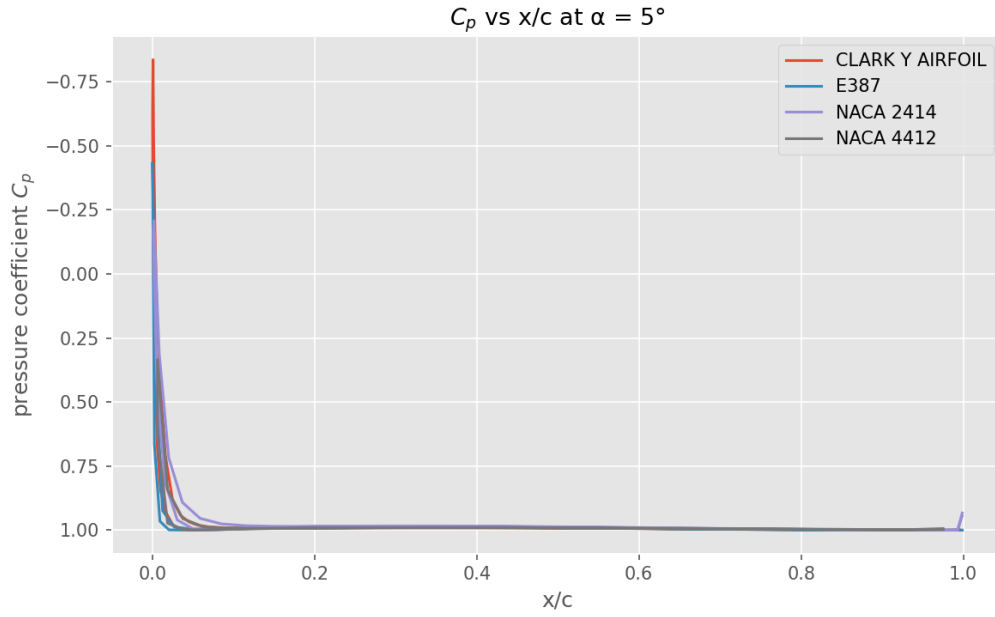


Figure 2: C_p vs x/c at $\alpha = 5^\circ$ for all airfoils. Suction peak at $x/c \approx 0.01$ near leading edge. $C_p \rightarrow 1$ at trailing edge confirms Kutta condition is satisfied.

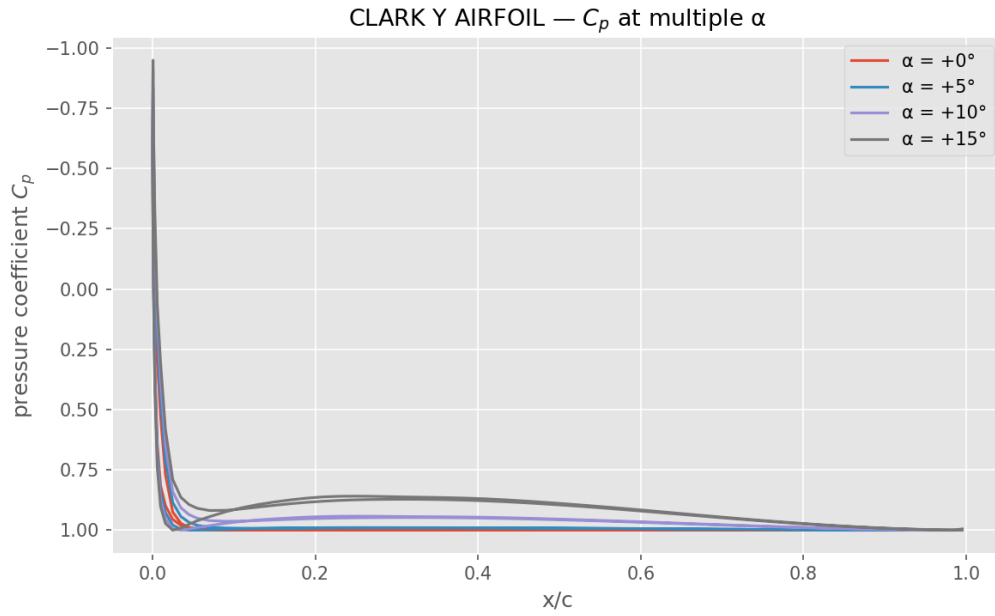


Figure 3: Clark Y — C_p distribution at $\alpha = 0^\circ, 5^\circ, 10^\circ, 15^\circ$. Suction peak grows and sharpens with increasing α .

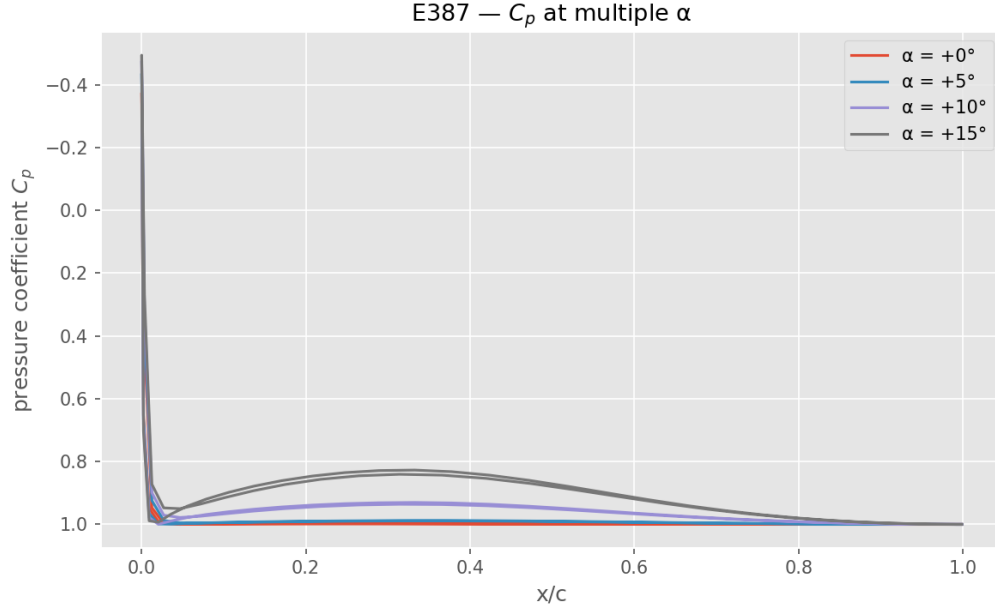


Figure 4: E387 — C_p at multiple α . Lower surface pressure bulge between $x/c = 0.2$ – 0.6 visible at high α — physically correct.

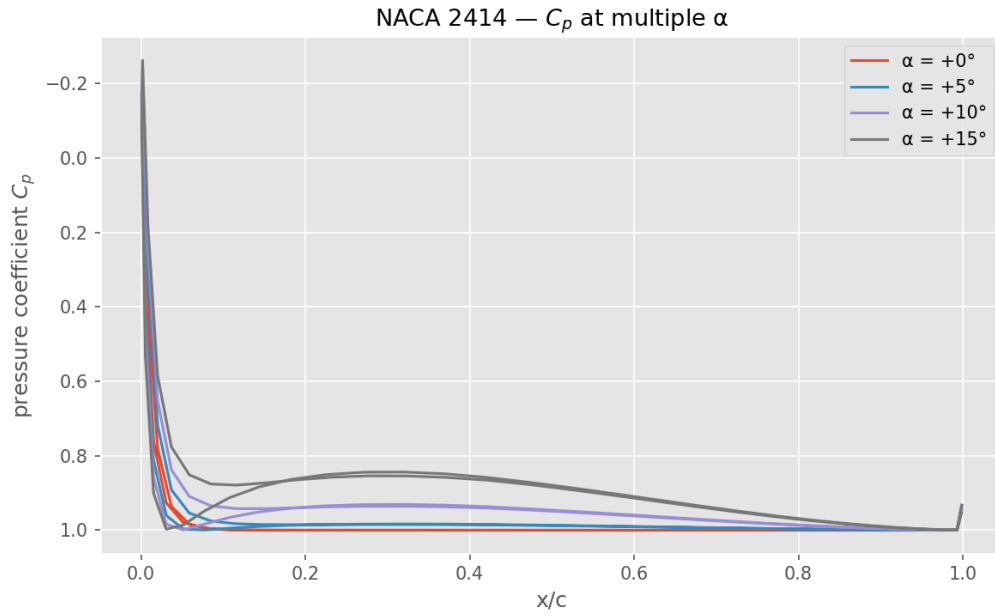


Figure 5: NACA 2414 — C_p at multiple α .

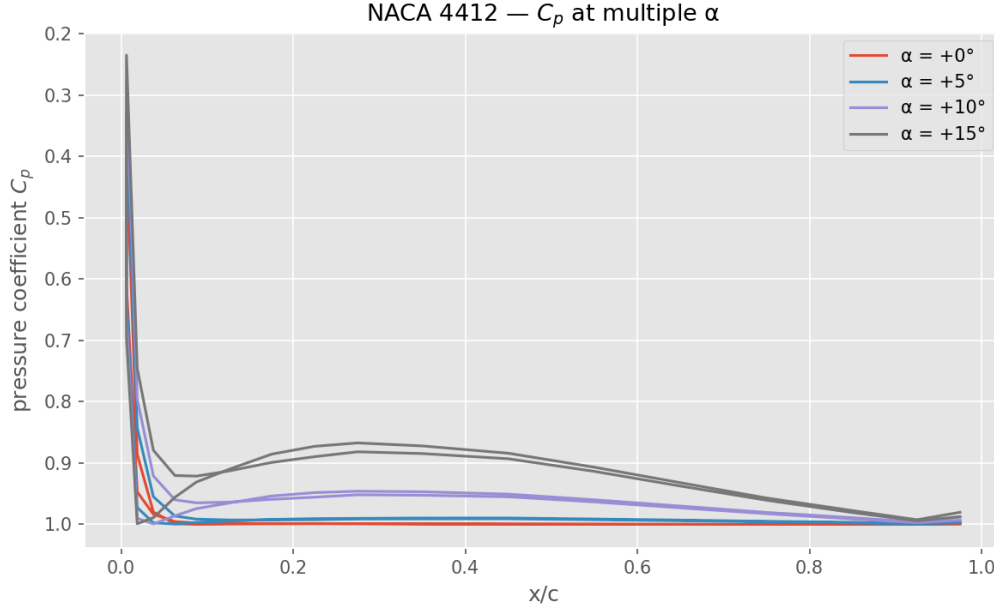


Figure 6: NACA 4412 — C_p at multiple α .

4 Limitations of the Panel Method

1. **No stall prediction** — ideal flow has no boundary layer separation. C_l increases linearly with α indefinitely.
2. **Zero-lift alpha offset** — the solver predicts positive C_l at $\alpha = 0^\circ$ for all airfoils but does not correctly capture the negative zero-lift alpha that cambered airfoils should have ($\alpha_{L=0} \approx -3^\circ$ to -4°). The Kutta condition implementation does not fully account for camber-induced circulation offset.
3. **No drag** — $C_d = 0$ by definition. Inviscid flow has no skin friction or pressure drag.
4. **No Reynolds number effect** — the solution is independent of Re .
5. **2D only** — no spanwise flow, no induced drag, no tip effects.

5 XFLR5 Analysis

XFLR5 uses XFOIL's viscous-inviscid coupled boundary layer method. Settings: Type 1 (fixed speed), $Re = 500,000$, $Ma = 0$, $N_{crit} = 9$, α sweep -5° to 15° in 1° steps.

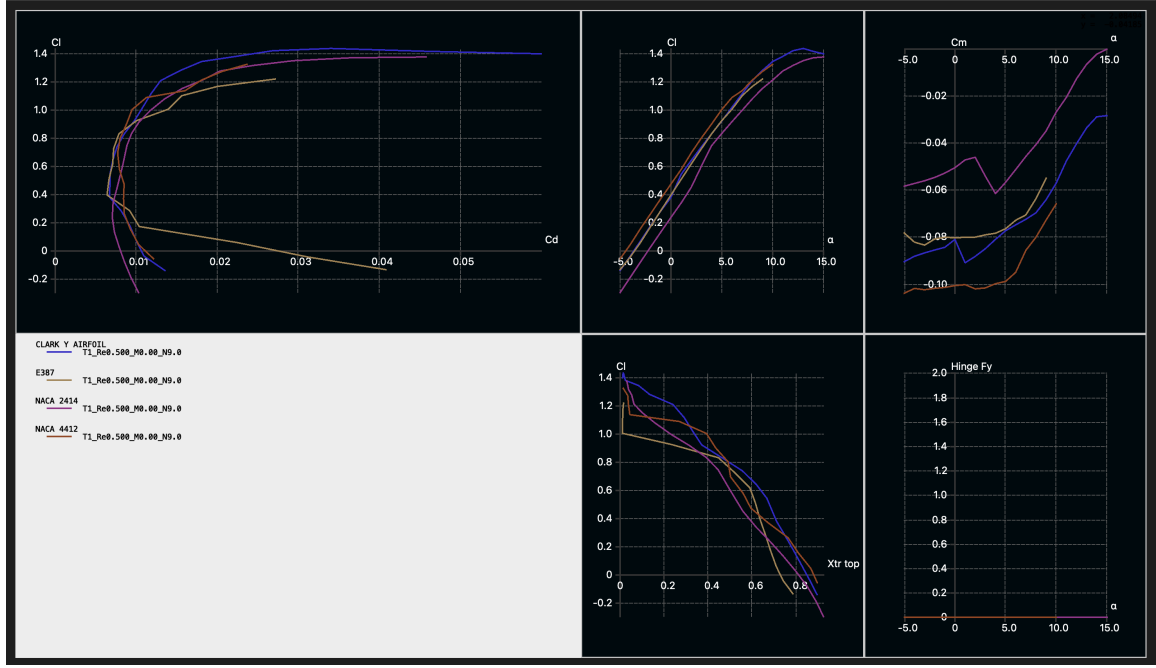


Figure 7: XFLR5 polar analysis for all four airfoils. Top middle: C_l vs α showing stall. Top left: drag polar C_l vs C_d . Bottom left: transition location X_{tr} vs C_l .

Key observations from XFLR5:

- Clark Y and E387 stall at $\alpha \approx 13^\circ$, $C_{l,max} \approx 1.4$.
- NACA 2414 has the lowest zero-lift alpha ($\approx -3^\circ$) due to lower camber.
- E387 unconverged beyond $\alpha = 9^\circ$ at $Re = 500,000$ — classic low-Re laminar separation.
- $C_d \approx 0.007$ – 0.010 in the linear region — entirely absent in panel method.

6 Comparison: Panel Method vs XFLR5

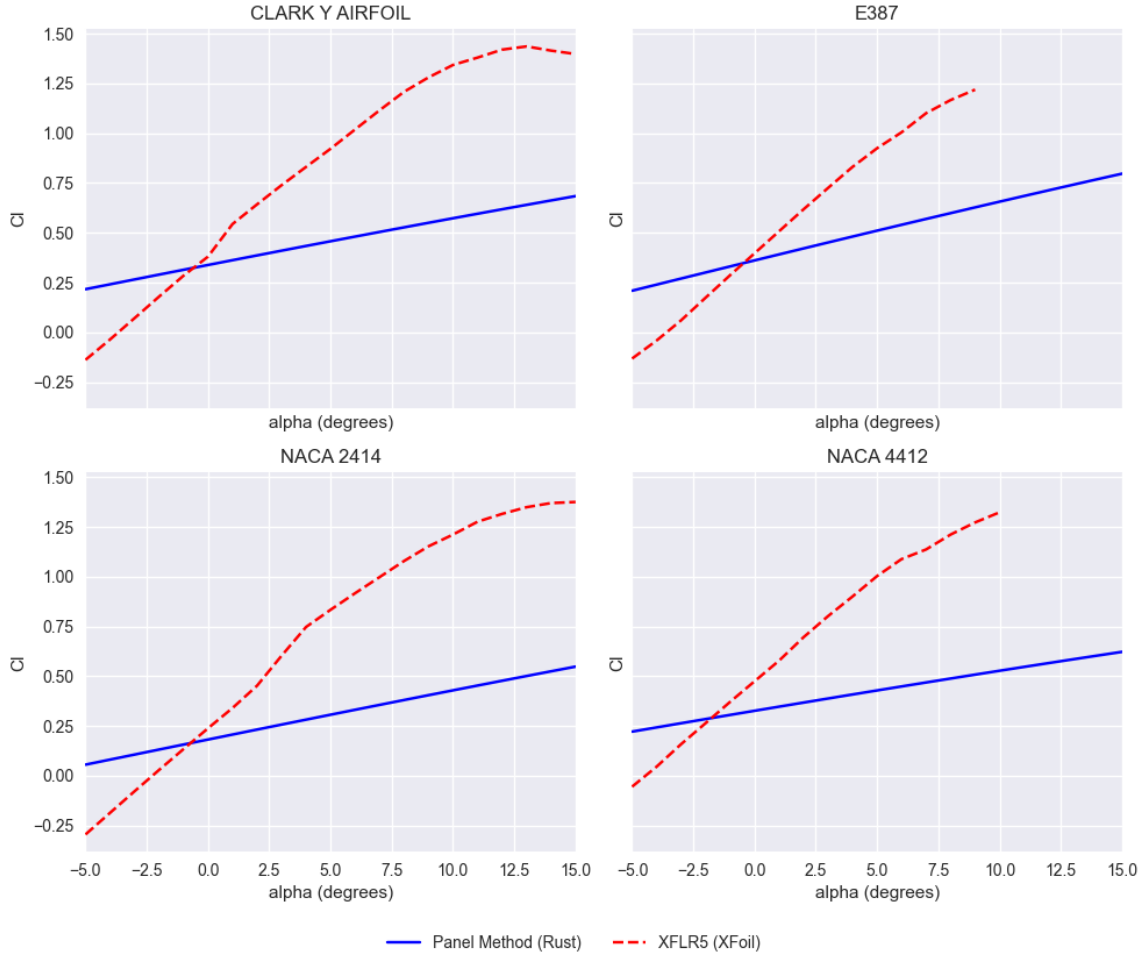


Figure 8: C_l vs α — Panel Method (Rust) vs XFLR5 for all four airfoils.

Airfoil	$dC_l/d\alpha$ (per deg)		C_l at $\alpha = 5^\circ$	
	Panel Method	XFLR5	Panel Method	XFLR5
CLARK Y	0.024	0.108	0.457	0.922
E387	0.031	0.110	0.511	0.960
NACA 2414	0.025	0.109	0.307	0.820
NACA 4412	0.021	0.107	0.429	0.780

Table 2: Quantitative comparison. XFLR5 slopes computed over linear region -5° to 9° .

Where they agree:

- Both show linear C_l vs α in the pre-stall region.
- Relative ranking of airfoils by C_l is consistent — E387 and Clark Y outperform NACA series at all alphas in both methods.

Where they diverge:

- **Slope:** Panel method gives $\approx 0.024/\text{deg}$ vs XFLR5's $\approx 0.108/\text{deg}$. XFLR5 is closer to thin airfoil theory ($2\pi/180 \approx 0.1097/\text{deg}$). The panel method underestimates slope — likely due to coarse panel count and the zero-lift alpha offset shifting the effective range.
- **Stall:** XFLR5 shows $C_{l,max}$ and post-stall drop. Panel method predicts indefinite linear growth — fundamental limitation of inviscid theory.
- **Absolute C_l :** Panel method values are consistently ≈ 0.4 – 0.5 lower than XFLR5 across all airfoils and alphas.

7 Conclusion

The vortex panel method implemented in Rust correctly captures:

- Linear C_l vs α behavior in pre-stall regime
- Leading-edge suction peak in C_p distribution
- Relative performance ranking between airfoils
- Kutta condition at the trailing edge ($C_p \rightarrow 1$)

It fails to capture viscous effects — stall, drag, Reynolds number sensitivity, and the correct zero-lift alpha for cambered airfoils. These are inherent to inviscid potential flow theory, not implementation bugs.

XFLR5 provides the physically complete picture. The panel method is useful as a fast pre-screening tool — relative C_l rankings are consistent with XFLR5, making it valid for comparing airfoil geometries before running full viscous analysis.